Candidate Seat No: _______________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
First Semester of B.Tech. Theory Examination(All Branches)
April-May 2018
MA101 Engineering Mathematics I

Date: 10/05/2018 (Thursday) **Time:01:30 pm to 04:30 pm** **Maximum Marks: 70**

Instructions:

i. The question paper contains two sections.
ii. Section I and II must be attempted in separate answer sheets.
iii. Figures to the right indicate marks.
iv. Make suitable assumptions and draw neat figure where it is required.
v. All notations and terminologies are standard.

Section-I

Q-1 **Choose the correct answer from the given options in the following:** **[07]**

1) For a Geometric Progression $5-\frac{10}{3}+\frac{20}{9}-\frac{40}{27}+\ldots$, the value of ratio of any two consecutive terms is___________. **01**

(a) $-\frac{2}{3}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{4}{9}$ **(d)** $-\frac{4}{9}$

2) If $\lim_{n\to\infty} u_n \neq 0$ for an infinite series $\sum_{n=1}^{\infty} u_n$ then the series is_________. **01**

(a) oscillating **(b)** convergent **(c)** divergent **(d)** alternating

3) If $z = 2+3i$ then the value of modulus of z is_________. **01**

(a) $-\sqrt{13}$ **(b)** $\sqrt{13}$ **(c)** $-\sqrt{5}$ **(d)** $\sqrt{5}$

4) If $z = 1+i$, then the value of $\arg(z)$ is__________. **01**

(a) $\frac{\pi}{4}$ **(b)** $-\frac{\pi}{4}$ **(c)** $\frac{\pi}{2}$ **(d)** $-\frac{\pi}{2}$

5) For a system of linear equations $AX = 0$, if rank$(A) =$number of unknowns in the system then the system has_______. **01**

(a) infinite solution **(b)** no solution **(c)** trivial solution **(d)** can not say anything

6) If two rows of matrix A are same, then the value of determinant of A is________. **01**

(a) 2 **(b)** -1 **(c)** 1 **(d)** 0

7) If nullity of a square matrix A of order 3 is 2, then the rank of A is_________. **01**

(a) 0 **(b)** 1 **(c)** 2 **(d)** 3

Q-2 Attempt any Four. **[16]**

1) Show that an infinite series $\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\ldots+\frac{1}{n\cdot(n+1)}+\ldots$ is convergent. **04**

2) Find the roots of $\sin z=0$. **04**

3) Find the real and imaginary part of $(1-i)^{152}$. **04**

4) Find the inverse of $A=\begin{bmatrix}1 & 0 & 0\\ -2 & 1 & 0\\ 0 & -1 & 5\end{bmatrix}$ by Gauss-Jordan method, if exist. **04**

5) Obtain the rank and nullity of a matrix $\begin{bmatrix}0 & 2 & 1\\ 5 & 2 & 3\\ 1 & 1 & 0\end{bmatrix}$ by converting it into row echelon form. **04**

6) Find the value of λ and μ so that the system of linear equations $2x+y+2z=5$; $x+2y+3z=6$; $x+2y+\lambda z=\mu$ has no solution. **04**

Q-3 (a) Attempt any Two. **[06]**

1) Obtain the interval of convergence of $\sum_{n=1}^{\infty}\frac{x^n}{n+1}$. **03**

2) Check the convergence of $\sum_{n=2}^{\infty}\log n$. **03**

3) Find rank of a matrix A using minors, where $A=\begin{bmatrix}-2 & -4\\ 3 & 6\end{bmatrix}$. **03**

Q-3 (b) State De-Moivre's formula and using it simplify **[06]**

$$\frac{(\cos 2\theta+i\sin 2\theta)^{2/3}(\cos\theta-i\sin\theta)^2}{(\cos 3\theta-i\sin 3\theta)^2(\cos 5\theta-i\sin 5\theta)^{1/3}}.$$

OR

Q-3 (b) Determine all distinct roots of $z^5=1$. **[06]**

Section-II

Q-4 Choose the correct answer from the given options in the following: **[07]**

1) The Maclaurin's series expansion for e^x is________. **01**

(a) $1+x+\frac{x^2}{2}+\frac{x^3}{3}+...$ **(b)** $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+...$

(c) $1-x-\frac{x^2}{2!}-\frac{x^3}{3!}-...$ **(d)** $1+x-\frac{x^2}{2}+\frac{x^3}{3}-...$

2) The value of $\lim_{n\to\infty} n^{\frac{1}{n}}$ is________. **01**

(a) 0 **(b)** ∞ **(c)** $-\infty$ **(d)** 1

3) If $f(x,y)=0$, then $\frac{dy}{dx}$ is__________. **01**

(a) $\frac{\partial f/\partial x}{\partial f/\partial y}$ **(b)** $\frac{\partial f/\partial y}{\partial f/\partial x}$ **(c)** $-\frac{\partial f/\partial y}{\partial f/\partial x}$ **(d)** $-\frac{\partial f/\partial x}{\partial f/\partial y}$

4) If $u=x^2+y^2+z^2$ be such that $xu_x+yu_y+zu_z=\lambda u$, then λ is________. **01**

(a) 1 **(b)** 2 **(c)** 3 **(d)** None of these

5) To obtain the maximum value of a function $f(x,y,z)$ subject to the condition $\phi(x,y,z)=0$ one may use_________. **01**

(a) Lagrange's multiplier method **(b)** Euler's method
(c) Newton's method **(d)** Bessel's method

6) The saddle point for the function x^3-y^2-3x is________. **01**

(a) (0,1) **(b)** (1,0) **(c)** (1,1) **(d)** (0,0)

7) If $x=r\cos\theta,\ y=r\sin\theta$, then the value of $\frac{\partial(r,\theta)}{\partial(x,y)}$ is________. **01**

(a) r **(b)** θ **(c)** $\frac{1}{r}$ **(d)** $\frac{1}{\theta}$

Q-5 Attempt any Four. **[16]**

1) Expand x^3-3x^2+4x+3 in powers of $(x-2)$ by using Taylor's series expansion. **04**

2) Using Euler's theorem find the value of $(i)\ x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}$ and $(ii)\ x^2\frac{\partial^2 u}{\partial x^2}+2xy\frac{\partial^2 u}{\partial x\partial y}+\frac{\partial^2 u}{\partial y^2}$, where $u=x^3\tan^{-1}\left(\frac{x^3+y^3}{xy^2}\right)$. **04**

3) Obtain the maximum value of $f = x^2y^3z^4$ subject to the condition $x+y+z=5$ using Lagrange's method of undetermined multiplier. **04**

4) If $u = x^2 + y^2 + z^2$, where $x = e^t$, $y = e^t \sin t$, $z = e^t \cos t$ find $\frac{du}{dt}$. **04**

5) If $u = \log(x^2 + y^2) + \tan^{-1}\left(\frac{y}{x}\right)$,then show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. **04**

6) Evaluate $\lim_{x\to 0}\left[\frac{1}{x^2} - \frac{1}{\sin^2 x}\right]$ using L'Hopital Rule. **04**

Q-6 **(a) Attempt any Two.** **[06]**

1) Find the extreme values of $x^2 + y^2 + 6x + 12$. **03**

2) If $u = x^2 - y^2$ and $v = 2xy$, then find $\frac{\partial(u,v)}{\partial(x,y)}$. **03**

3) Find the percentage error in calculating the area of a rectangle when an error of 3% is made in measuring each of its sides. **03**

Q-6 **(b)** If $y = \tan^{-1} x$,then prove that $(1+x^2)y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$. **[06]**

OR

Q-6 **(b) (i)** Find the n^{th} derivative of $y = \frac{1}{x^2 - 6x + 8}$. **[06]**

(ii) Find the n^{th} derivative of $y = \frac{x+2}{x+1} + \log\left(\frac{x+2}{x+1}\right)$.

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